CSE4006 – DL Assignment – 2 – Submission End Date: 02.04.26.

1. A medical imaging research team wants to classify chest X-ray images into normal and pneumonia categories. Due to limited labeled data, they decide to use a pre-trained VGG16 model.
a) Explain the concept of Transfer Learning.
b) Describe the two approaches of Transfer Learning: Feature Extraction, Fine-Tuning
c) Which approach is more suitable when the dataset is very small? Justify your answer.

2. A company is building an AI-powered real-time language translation system that processes long sentences.
a) Compare LSTM and GRU architectures in terms of, number of gates, computational complexity, memory usage, training efficiency.
b) Which architecture would you recommend for real-time applications? Justify your answer.

3. A speech-to-text system needs to interpret words based on both past and future context.
a) Explain how a standard LSTM processes sequences.
b) Explain how a Bidirectional LSTM processes sequential data.
c) Why do Bidirectional LSTMs improve performance in NLP tasks?

4. Deep learning models often suffer from overfitting when training data is limited.
a) Explain the concept of Data Augmentation.
b) Describe techniques used to handle the following invariances: Translational invariance, Rotational invariance, Scale invariance
c) Explain how data augmentation improves model generalization.

5. Recurrent Neural Networks come in different architectures depending on the type of sequence task. Explain the various RNN architectures with suitable applications.

6. Deep CNN architectures introduced new modules to improve performance.
a) Explain the **Inception Module in GoogleNet**.
b) What is the role of **Auxiliary Classifiers** in GoogleNet?
c) Explain the concept of a **Residual Block in ResNet**.
d) How do **skip connections help overcome the vanishing gradient problem**?

7. Consider a CNN with the following layers:
- Convolution layer: **16 filters**, size **3×3**, input channels **3**
- Convolution layer: **32 filters**, size **3×3**, input channels **16**
- Fully connected layer: **128 input units**, **64 output units**
- Batch normalization applied after each convolution layer

Calculate the **total number of trainable parameters in the network**.

8. Overfitting is a common problem in deep learning.
a) Define **Regularization** in deep learning.
b) Explain the working of **Dropout**.
c) Explain the concept of **Early Stopping**.
d) Discuss how these techniques help improve **model generalization**.

9. Training deep neural networks is often slow due to internal covariate shift.
a) Explain the concept of **Batch Normalization**.
b) Write the **mathematical equations** used in Batch Normalization.
c) Explain how Batch Normalization helps:

- accelerate training
- stabilize learning
- allow higher learning rates.

10. Consider an **RNN with 3 time steps**.
Given Inputs, $x_1 = 1, x_2 = 2, x_3 = 1$, Targets, $y_1 = 0.5, y_2 = 0.8, y_3 = 0.3$, Initial hidden state $h_0 = 0.5$, Weights, $W_{xh} = 0.5$, $W_{hh} = 0.2$, $W_{hy} = 0.9$, Activation functions: Hidden layer: **tanh,** Output layer: **sigmoid.** Loss function (MSE):

$$\boldsymbol{L = \frac{1}{2}\sum_{t=1}^{3}(y_t - \hat{y}_t)^2}$$

a) Perform the **forward pass** and compute:

- h_1, h_2, h_3
- $\hat{y}_1, \hat{y}_2, \hat{y}_3$

b) Compute the **total loss**.
c) Using **Backpropagation Through Time**, compute the gradients: $\frac{\partial L}{\partial W_{hy}}, \frac{\partial L}{\partial W_{xh}}, \frac{\partial L}{\partial W_{hh}}$.
d) Update the weights using **learning rate = 0.1**.

11. Adaptive gradient-based optimization algorithms have significantly advanced the training of deep neural networks by addressing the limitations of vanilla stochastic gradient descent. With respect to **AdaGrad**, **RMSProp**, and **Adam**:
(a) Derive the parameter update rule for each optimizer, clearly defining all notations including accumulated gradient terms, decay rates, and bias-correction factors.
(b) Analyze the key limitation of AdaGrad that motivated the development of RMSProp, and explain how RMSProp's exponential moving average of squared gradients resolves this issue.
(c) Demonstrate how Adam synthesizes the principles of both RMSProp and momentum-based optimization, and justify the necessity of bias correction in its first and second moment estimates.
(d) Compare the convergence behavior of all three optimizers on non-stationary and sparse gradient problems, supporting your analysis with mathematical justification.